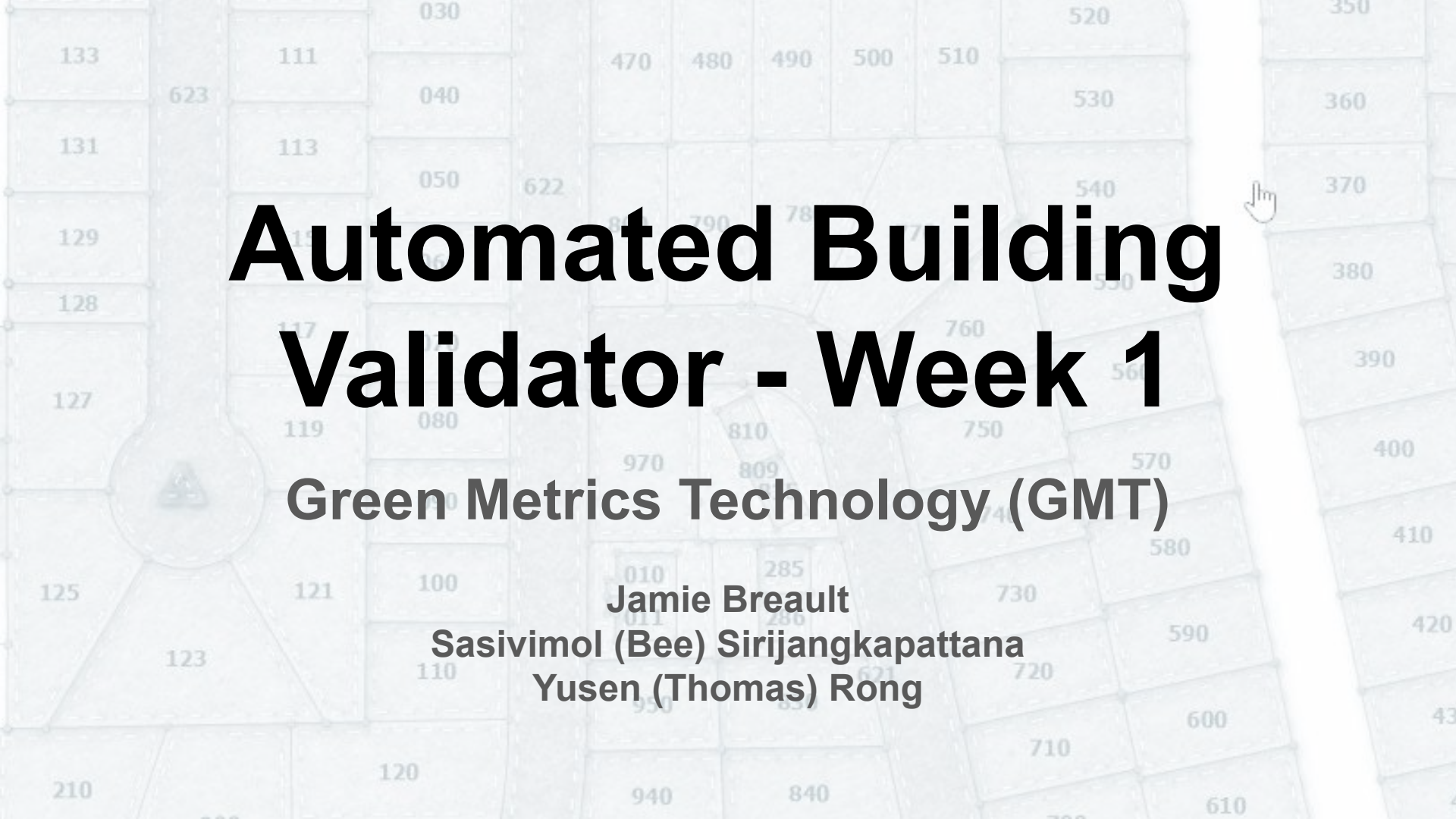


The background is a grayscale map of a city grid, showing various street blocks and individual lots. Each lot is labeled with a four-digit parcel number. A hand cursor icon is positioned on the right side of the map, pointing towards the text.

Automated Building Validator - Week 1

Green Metrics Technology (GMT)

Jamie Breault

Sasivimol (Bee) Sirijangkapattana

Yusen (Thomas) Rong

Project Overview

I want to build a house in BC, but I don't know where I can legally build

Research Questions:

1. How to discover bylaw sources across jurisdictions?
2. How to retrieve rule-relevant content from fragmented sources?
3. How to determine clashes between bylaws and building plans?
4. How to maintain end-to-end provenance linking outputs to the source?

Team Operations

- Coding Lead - Thomas
 - Organize git repo, responsible for high-level architecture
 - Reporting Lead - Bee
 - Maintain documentation, meeting notes
 - Facilitator - Jamie
 - Ensure project is on track, partner communication
-
- Weekly progress summary meeting with GMT every Monday
 - Weekly in-person team meetings after DATA 599 class
 - Daily check-ins on Discord, and ad-hoc discussions
 - “Blocker Reports” submitted to GMT as team, at least weekly

Progress Update

- Gained housing development domain knowledge
 - Setback distances, conditions, and exceptions
 - Building height definitions
- Investigated tools and methods for each composite part
 - Web scraping with Python to find datasets
 - RAG to find rules in bylaw documents
 - GIS to determine if a building is permitted geometrically
- Produced a project proposal that outlines the project motivation, background, data sources, objectives, timeline, and more

Challenges

- Working with fragment datasets
 - Need to explore web scraping options in depth, every municipality is unique
- Working with unfamiliar methods and tools (RAG and GIS)
 - Investigating methods to ensure feasibility and compatibility with the project
- Understanding the scope of the project and how to approach it
 - Discussed in the weekly meetings, still a work in progress

Next Steps and Timeline

- Divided the work into three subproblems, one team member will focus on each
- The output of one subproblem can be fed into the input of the next
- Data provenance and a UI is an extended goal



Week	Dates	Data Source Discovery (Sasivimol)	Rule Discovery (Sasivimol)	Clash Detection (Jamie)
0	April 27 to April 30	- Introduce UBC team to Google Maps API - Gain housing development data from Google Maps		
1	May 4 to May 7	- Complete project setup and data collection methods, tools, and timeline		
2	May 11 to May 14	- Start parcel shapefile processing and get to 50% complete		- Install QGIS and familiarity with parcel GIS data - Fetch zoning type by address or PID given the needed parcel file (for Rule Retrieval)
3	May 18 to May 21	- Continue parcel shapefile processing and get to 75% complete		- Complete solution to define the legal buildable area of a parcel using geometric rules
4	May 25 to May 28	- Continue parcel shapefile processing and get to 90% complete		- Complete solution to determine whether the building fits in the legal buildable area, get to 75% complete
5	June 1 to June 4	- Continue parcel shapefile processing and get to 95% complete		- Complete solution to determine whether the building fits in the legal buildable area, get to 90% complete
6	June 8 to June 11	- Complete parcel shapefile processing and get to 100% complete	- Complete rule discovery and get to 100% complete	- Complete solution to determine whether the building fits in the legal buildable area, get to 100% complete
7	June 15 to June 18	- Test accuracy of zoning bylaw scraper and refine	- Test accuracy of rule retrieval and refine	- Test accuracy of clash detection and refine
8	June 22 to June 23	- Prepare final presentation	- Complete final report	

Table 1: Weekly goals for all team members.

Client Communications

- Weekly Monday meeting with GMT
 - Discussed our project proposal
 - GMT expressed concern over scope
 - GMT wanted us to investigate rule attributes more
 - Once complete, a new timeline with reduced scope will be discussed
- Daily ad-hoc Discord discussions
 - GMT suggested introducing a rule attribute for rule-source fidelity
 - GMT encouraged us to use the question and blocker protocol they outlined

Updated Next Steps

- Determine which types of rules exist for single laneway homes
 - What attributes can a rule have?
 - What values can each attribute have?
- Each team member will investigate the zoning bylaws of one municipality
 - Bee has Vancouver
 - Thomas has Surrey
 - Jamie has Burnaby
- Identify common rules
- Meet with GMT and establish new timeline

```
{ "rule":  
  { "municipality": "Vancouver",  
    "key": "max_height",  
    "code": "101.5.1",  
    "type": "Max",  
    "value": 10,  
    "condition": "roof is flat",  
    "penalty": "straight to jail"  
  }  
}
```

Contributions

Team Member	Primary Tasks	Hours
Jamie	1. Researched GIS 2. Wrote proposal - Motivation - Timeline - Deliverables - Editing & Formatting	40
Bee	1. Researched web scraping tools 2. Wrote proposal - Problem Overview - Background - Objectives - Conclusion - References	33
Thomas	1. Researched RAG 2. Wrote proposal - Data Sources	32